

Network Architecture Components - Detailed Explanation

This document explains the components of a secure enterprise network architecture and the role of each element in protecting systems, data, and users.

Component	Description / Role
Internet	External untrusted network used by users to access services.
External Zone	Untrusted buffer zone between internet and internal network.
Customers	External users who access the system and services.
External Firewall	Filters and blocks unauthorized traffic from the internet.
WAF	Protects web applications from attacks like SQL Injection and XSS.
IPS	Monitors traffic and blocks malicious activities automatically.
IDS	Detects suspicious activities and generates alerts for security teams.
DMZ	Isolated zone for hosting public-facing servers securely.
Web Server	Handles HTTP requests and serves web application content.
Internal Firewall	Controls traffic between DMZ and internal network.
Internal API	Secure communication layer between internal systems.
App Server	Processes business logic and application requests.
Database Server	Manages database engine and data operations.
Database	Stores and organizes system and user data securely.
VPN Gateway	Provides encrypted secure remote access to internal network.
Remote Employees	Users accessing the network from outside locations securely.
Internal Employees	Authorized users working inside the organization.
SIEM	Collects and analyzes logs for real-time threat detection.
SBOM	Inventory of software components for vulnerability tracking.
AES-256 Encryption	Strong encryption standard used to protect sensitive data.
DevSecOps	Integrates security into software development lifecycle.
Trusted Zone	Secure internal network area containing critical systems.